

Tianfang Chang

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Summary

Software Engineer with experience in C++, Python, and Rust systems development across distributed networking research and end-to-end desktop software engineering. Built multithreaded, performance-sensitive systems involving lock-free concurrency, SQLite-backed data pipelines, native Windows audio integration, and React/Tauri desktop applications. Interested in Software Engineer roles focused on systems, backend infrastructure, and performance-critical applications.

Skills

Languages: Rust, C++, Python, TypeScript, SQL, Bash/Shell

Systems & Backend: Multithreading, Lock-free Concurrency, SQLite, r2d2, IPC, Windows COM, WASAPI, ASIO, Data Ingestion Pipelines

Frontend & Tools: React, Tauri, Zustand, TanStack Virtual, Docker, GitHub Actions, CI/CD, Git, Linux, Windows, ns-3

Experience

Software Research Engineer – University of Victoria Jul 2025 – Nov 2025

- Built a deep-learning-based PyTorch model for satellite signal-strength prediction from geospatial data.

Software Research Engineer – University of Victoria Sep 2021 – Aug 2024

- Prototyped protocol components for next-generation networks in C++/Python using Network Simulator 3 (ns-3), improving efficiency and enabling new applications.

Teaching Assistant, ECE 363 – University of Victoria Jan 2022 – Apr 2023

- Mentored 40+ students in TCP/IP network simulation and performance measurement projects using Linux network tools and Python automation.

Software Engineer Intern – Institute of Computing Technology, CAS Aug 2019 – Dec 2019

- Built a server resource scheduler using Python heuristic algorithms (ant colony, genetic search), improving cluster utilization by ~18% while reducing average response time under mixed workloads.

Projects

FFAudio – Bit-Perfect Desktop Audio Engine GitHub

Stack: Rust, Tauri, React/TypeScript, SQLite

- Developed a native multithreaded audio engine in Rust, interfacing directly with Windows WASAPI/ASIO endpoints to bypass OS-level mixer resampling for bit-perfect playback.
- Implemented a lock-free, zero-allocation ring buffer (`rttrb`) to synchronize decoding and real-time audio callback threads without mutex contention or playback dropouts.
- Built a concurrent local library indexing pipeline with Rayon and SQLite using separate read/write pools via `r2d2`, including CJK metadata recovery with `chardetng`.
- Designed a desktop client with React, TypeScript, and Tauri IPC, including virtualized large-list rendering, deterministic state management with Zustand, bilingual i18n support, and CI workflows using GitHub Actions.

Delay-guaranteed Routing Protocol GitHub

Stack: C++, Python, ns-3, Linux, TCP/IP

- Designed and developed DDR, a distributed traffic-aware routing protocol providing packet-level delay guarantees for time-sensitive applications.
- Built reproducible ns-3 experiments on ISP-grade topologies and evaluated DDR against ECMP, LFID, and DGRP under mice, elephant, and bursty traffic workloads.

Multipath Routing Compatible TCP GitHub

Stack: C++, Python, ns-3, Linux, TCP/IP

- Implemented an RTT-aware FastRTO mechanism to decouple packet reordering from loss, reducing false fast retransmits on multipath routes.
- Built reproducible ns-3/Ubuntu benchmarks against NewReno, CUBIC, and BBR on OSPF single-path and ECMP multipath topologies.

Education

University of Victoria Master of Applied Science, Electrical and Computer Engineering CNLab (Computer Network Lab)	Sep 2021 – Aug 2024
Beijing University of Posts and Telecommunications Bachelor of Engineering, Telecommunication Engineering	Sep 2015 – Jun 2020

Publications

Multipath Routing Compatible Congestion Control Tianfang Chang, Lin Cai 2024 IEEE 100th Vehicular Technology Conference	VTC 2024
DDR: A Deadline-Driven Routing Protocol for Delay Guaranteed Service Pu Yang, Tianfang Chang, Lin Cai 2024 IEEE International Conference on Computer Communications	INFOCOM 2024
DGR: Delay-Guaranteed Routing Protocol Pu Yang, Lin Cai, Tianfang Chang 2023 IEEE 20th International Conference on Mobile Ad Hoc and Smart Systems	MASS 2023

Awards & Leadership

Fellowship , University of Victoria Graduate Student	2021 – 2024
Finalist , IEEE VTS Storytelling Competition	2023
Finalist , IEEE 2023 Workshop on Ubiquitous Poster Competition	2023
Scholarship , University of Victoria Graduate Student	2022

Community Activity

Paper Reviewer IEEE INFOCOM, IEEE Network Magazine, IEEE TWC, IEEE TMC	2022 – 2024
Volunteer IEEE 2023 Workshop on Ubiquitous	2023
Volunteer Explore UVic Event	2023